MILITARY SPECIFICATION

BURS, BONE SURGERY (Corrosion-Resisting Steel)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers general requirements for steel, bone surgery burs used in DoD Deployable Medical Systems (DMS).

1.2 Classification.

1.2.1 Head. Corrosion-resisting steel, bone surgery burs shall be of round head with the following head diameters.

Diameter (mm)
2.4
3.2
4.0
4.8
5.5
6.3

1.2.2 Overall Length. Corrosion resisting steel, bone surgery burs shall be of the following overall length.

Length (mm)
44
55
73

AMSC N/A FSC 6515

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

GG-I-526	- Instruments, Dental and Surgical; General Specifications.
PPP-B-601	- Boxes, Wood, Cleated-plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

Military

MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-B-22191	- Barrier Material, Transparent, Flexible, Heat Sealable.
MIL-P-116	- Preservation, Methods of.
MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Federal

FED-STD-151	- Metals: Test Methods.

Military

MIL-STD-105	- Sampling Procedures and Table for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-147	- Palletized Unit Loads.
MIL-STD-794	- Parts and Equipment, Procedures for Packaging of.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2. Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 18 Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.
E 92 Standard Test Method for Vickers Hardness of Metallic Materials.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

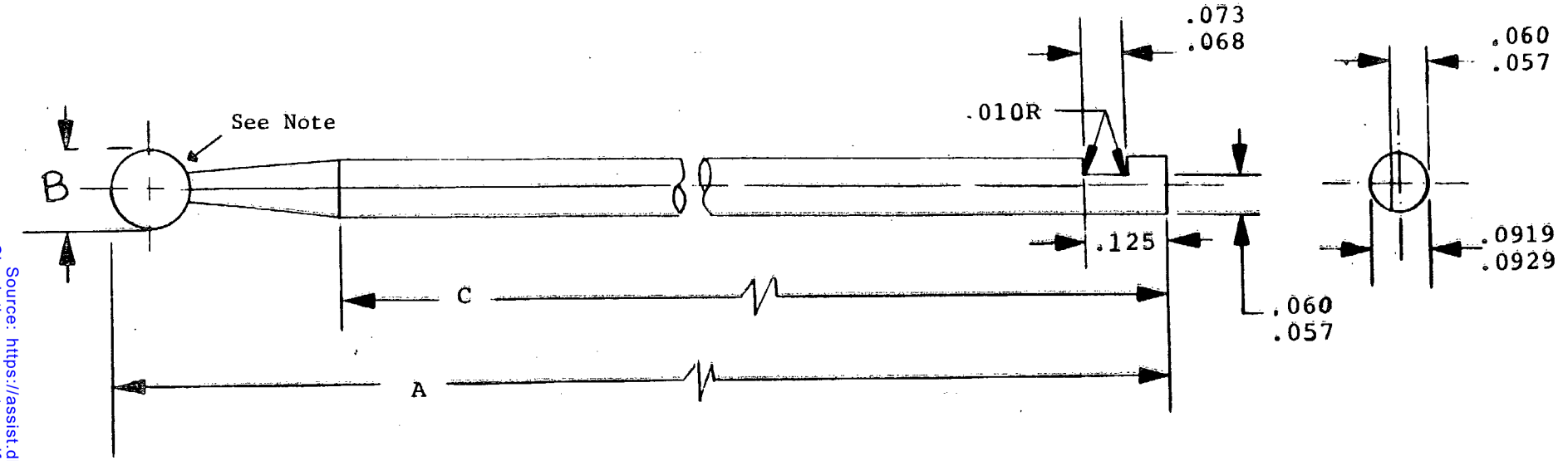
3.1 General requirements. The burs shall be in accordance with applicable requirements of GG-I-526 with the options and additional requirements specified herein.

3.2 Material. The burs shall be fabricated of type 416 corrosion-resisting steel.

3.3. Hardness. The burs shall be properly hardened and passivated. After proper stress relieving the burs shall have a Rockwell hardness of C40 to C45.

3.4 Style, design and construction. The burs shall be of round head and one piece construction. The burs shall be compatible for use with the appropriate bur guard component of PNEUMATIC BONE DRILLS, National Stock Numbers(NSNs) 6515-01-167-2619; 6515-01-167-2620 and 6515-01-167-2621. The proximal end of the bur shall be of J-notched configuration. The bur shall conform to figure 1 for design and dimensions for the applicable NSN and specification designation (see 1.2).

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NOTE: Burs with head diameters 3.2 mm or less shall have a tapered neck. All other burs shall have a straight neck.

National Stock No.	A(mm)	B(mm)	C(mm)	No. of flutes
6515-01-166-7784	73	4.8	64	6
6515-01-166-7785	73	3.2	64	6
6515-01-166-7786	55	4.8	46	6
6515-01-166-7787	55	3.2	46	6
6515-01-166-7788	44	6.3	35	8
6515-01-166-9005	55	4.0	46	6
6515-01-166-9006	44	2.4	35	4
6515-01-167-2616	44	5.5	35	8
6515-01-167-2622	44	4.0	35	6
6515-01-167-2623	73	4.0	64	6
6515-01-167-3788	44	3.2	35	6
6515-01-167-3789	44	4.8	35	6

All dimensions in inches, except as noted.

FINISH	TOLERANCES
Machine Finish to be 64 Microinches unless otherwise specified	± 1/64 on Fractional Dims. ± .010 on Decimal Dims. Unless otherwise specified Angles ± 1/2°

Figure 1. Burs, Bone Surgery

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3.5 Runout. The total indicated runout of the burs shall not exceed .001 inch when measured as specified herein.

3.6 Autoclaving. The burs shall not show evidence of corrosion when autoclaved at 270°F ± 5°F and 27 ± 1 pound per square inch pressure for a minimum of 20 minutes.

3.7 Corrosion resistance. Corrosion tests, (copper sulfate and boil test), as specified in GG-I-526 shall apply.

3.8 Finish. The operative end shall have a honed finish with sharpened cutting edges for fast cutting. The neck and shank shall have a smooth finish.

3.9 Marking. Marking of the burs shall be optional. The use of a trademark is acceptable. The type of marking or placement of the marking shall not adversely affect or interfere with fit, function and operation of the bur with handpiece. The marking shall be distinguishable to the unaided eye.

3.10 Instructions. Adequate and accurate instructions shall be furnished with each unit on the use and care of the burs. Instructions shall include methods of sterilization and sanitization. The instructions shall also include the type of solutions, both suitable and unsuitable for cold sanitization of the burs.

3.11 Workmanship. The burs shall have uniformly cut and formed blades and shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of supplies to which such records relate.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the component or materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.1.4 Inspection lot. An inspection lot, for purposes of sampling and testing in accordance with MIL-STD-105, shall consist of burs of the same size, manufactured under essentially the same conditions. Each unit package of burs shall be permanently marked with a serial number, or a combination of letters and numbers, which shall refer to the manufacturer's records for that particular lot or batch.

4.2 Quality conformance inspection. The examinations required to assure conformance of those burs or lots of burs to be offered for acceptance are classified as specified in tables I and III (see 4.2.1, 4.3 and 4.3.3. The test required for acceptance are specified in table II of 4.2.2.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQLs) shall be as shown in table I. The unit of product for examination purposes shall be the individual bur.

TABLE I. Examination criteria

Type defect	Inspection level	AQL (Percent Defective)
Major A	II	1.5
Major B	S-1	2.5
Minor	II	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified in table II. The acceptance number shall be zero for all sample sizes. The unit of product for test purposes shall be the individual bur, unless otherwise stated in the table.

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TABLE II. Test criteria

Characteristic	Requirement	Test procedure	Inspection level
Chemical composition	3.2	4.4.1	see 4.2.3
Hardness	3.3	4.4.2	S-2
Copper sulfate test	3.7	4.4.3.1	S-2
Boil test	3.7	4.4.3.2	S-2
Autoclaving	3.6	4.4.4	S-1

4.2.3 Raw material. A sufficient quantity of each lot of raw material used in the manufacture of the burs shall be selected for chemical analysis.

4.2.4 Sample unit usage. Where feasible, the same burs shall be used for the determination of two or more test characteristics.

4.3 Examination. The burs shall be examined for the possible defects listed in table III. The examination shall not be limited to those defects shown in the table. Unless otherwise indicated in the table, the defect listed shall apply to all burs covered by this specification.

TABLE III. Classification of defects

Category	Defects
Critical:	None defined.
Major A:	
101	Bur size and shape not as specified.
102	Bur not of one piece construction.
103	Total indicated runout exceeds 0.001 inch.
104	Number of flutes not as specified for the applicable bur.
Major B:	
151	Dimensions not as specified.
Minor:	
201	Cutoff end on shank not as shown in applicable figure.
202	Burs not free from defects such as chipped blades, cracks, or pits.
203	Finish on bur not as specified.
204	Overall length of bur not within specification limits.
205	Instructions for use inaccurate, inadequate or missing.
206	Packaging and packing not as specified.
207	Marking on packaging or packing incorrect, illegible, incomplete or missing altogether.

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4.3.1 Dimensions. Examination of the dimensions shall be made with a toolmaker's microscope, or other instrument of equal accuracy. Manufacturer's gages may be used if properly calibrated checking standards are used in conjunction with the gages.

4.3.2 Runout. The bur shall be placed and held in a "V" block or precision chuck, and rotated slowly. A dial indicator sensitive to 0.0002 inch and equipped with a knife-edge foot, or a shadow-graph or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is used, the indicator shall be mounted with the shaft perpendicular to the axis of the bur, and so positioned that the foot contacts the bur neck at the point of smallest diameter. The maximum and minimum readings obtained during one complete 360-degree rotation of the bur shall be measured and recorded to the nearest 0.0005 inch. The difference between these two readings shall be taken as the total indicated runout.

4.3.3 Visual examination. Examination for possible bur defects not perceptible to the unaided eye shall be made using a 10X magnification.

4.4 Tests. Tests shall be conducted to determine compliance with the requirements of this specification (see 4.2.2).

4.4.1 Chemical composition. In the absence of certificates of quality for the raw material, chemical analysis shall be conducted in accordance with method 111 or 112 of FED-STD-151. In case of dispute, method 111 shall govern.

4.4.2 Hardness. Hardness shall be determined in accordance with ASTM Designation E18 (Rockwell) or E92 (Vickers or Diamond Pyramid).

4.4.3 Corrosion resistance.

4.4.3.1 Boil test. The bur shall be subjected to the boil test, as specified in GG-I-526.

4.4.3.2 Copper sulfate test. The bur shall be subjected to the copper sulfate test, as specified in GG-I-526.

4.4.4 Autoclaving. The test bur shall be cleaned in accordance with the instructions supplied by the manufacturer. The bur shall then be autoclaved at 270 $\pm$ 5°F and 27 $\pm$ 1 pounds per square inch pressure for a minimum of 20 minutes. After the bur is removed from the autoclave, the entire surface area of the bur shall be examined for evidence of corrosion.

4.5 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Burs shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Burs shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative applications. Preservatives shall not be used.

5.1.1.4 Unit package. Each bur shall be packaged in a rigid plastic container having a telescopic friction or plug type closure. The bur shall be cushioned or immobolized within the container in such a manner as to prevent damage to the operative end.

5.1.1.5 Intermediate package. Six unit packages containing burs of one size only shall be packaged in a snug fitting sealed bag constructed of barrier material conforming to Grade A of MIL-B-121 or Type II of MIL-B-22191.

5.1.2 Level C. The level C preservation for burs shall conform to MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Ninety-six intermediate packages containing burs of one size only shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

5.2.2 Level B. Ninety-six intermediate packages containing burs of one size only, shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

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5.2.3 Level C. The level C packing for burs shall conform to MIL-STD-794 requirements for this level.

5.2.4 Packaging variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 40 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 43 inches in height, 48 inches in width, and 40 inches in length, and shall conform to Type IV or Type V.

5.2.5.1 Levels A and B. Burs packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets conforming to MIL-STD-147.

5.2.5.2 Level C. Burs packed as specified in 5.2.3 shall be unitized as specified in 5.2.5.

5.3 Marking.

5.3.1 Levels A and B, C. Each unit pack, intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

6.1 Intended use. Burs covered by this specification are intended for use in medical war readiness assemblages and DoD Deployable Medical Systems.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition document should specify the following:

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a. Title, number, and date of this specification.
b. NSN (as applicable).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 National Stock Number (NSN) coverage. This specification covers the following items appearing in the Federal Supply Catalog:

NATIONAL STOCK NUMBER	ITEM IDENTIFICATION
6515-01-166-7784	BUR, BONE SURGERY, Fast cutting, Corrosion-Resisting Steel, 4.8 mm Head Diameter, 6 Flutes, 73 mm Long
6515-01-166-7785	BUR, BONE SURGERY, Fast Cutting, Corrosion-Resisting Steel, 3.2 mm Head Diameter, 6 Flutes, 73 mm Long
6515-01-166-7786	BUR, BONE SURGERY, Fast Cutting, Corrosion-Resisting Steel, 4.8 mm Head Diameter, 6 Flutes, 55 mm Long
6515-01-166-7787	BUR, BONE SURGERY, Fast Cutting, Corrosion-Resisting Steel, 3.2 mm Head Diameter, 6 Flutes, 55 mm Long
6515-01-166-7788	BUR, BONE SURGERY, Fast Cutting, Corrosion-Resisting Steel, 6.3 mm Head Diameter, 8 Flutes, 44 mm Long
6515-01-166-9005	BUR, BONE SURGERY, Fast Cutting, Corrosion-Resisting Steel, 4.0 mm Head Diameter, 6 Flutes, 55 mm Long
6515-01-166-9006	BUR, BONE SURGERY, Fast Cutting, Corrosion-Resisting Steel, 2.4 mm Head Diameter, 4 Flutes, 44 mm Long
6515-01-167-2616	BUR, BONE SURGERY, Fast Cutting, Corrosion-Resisting Steel, 5.5 mm Head Diameter, 8 Flutes, 44 mm Long
6515-01-167-2622	BUR, BONE SURGERY, Fast Cutting, Corrosion-Resisting Steel, 4.0 mm Head Diameter, 6 Flutes, 44 mm Long
6515-01-167-2623	BUR, BONE SURGERY, Fast Cutting, Corrosion-Resisting Steel, 4.0 mm Head Diameter, 6 Flutes, 73 mm Long
6515-01-167-3788	BUR, BONE SURGERY, Fast Cutting, Corrosion-Resisting Steel, 3.2 mm Head Diameter, 6 Flutes, 44 mm Long

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NATIONAL STOCK NUMBER	ITEM IDENTIFICATION
6515-01-167-3789	BUR, BONE SURGERY, Fast Cutting, Corrosion-Resisting Steel, 4.8 mm Head Diameter, 6 Flutes, 44 mm Long

MILITARY INTERESTS:

Custodians
- Army-MD
- Navy-MS
- Air Force-03

Review Activity:
- Army-MD
- Navy-MS
- Air Force-03

PREPARING ACTIVITY:
- DoD-MB

AGENT:
- DLA-DM

Civil Agency Coordinating Activity:

VA-OSS

Project No. 6515-3244P

Location: NCR

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-37965	Burs, Bone Surgery (Corrosion Resisting Steel)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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